

PROJECT NAME

PROJECT REPORT

*Submitted in partial fulfillment of the requirements for the award of the
Degree of **Bachelor of Technology in Electronics & Communication
Engineering of APJ Abdul Kalam Technological University***

By

YOUR NAME

(VI SEMESTER B.TECH, REG No. 13123XXX)

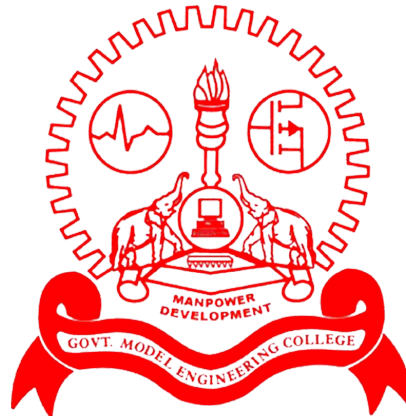


MODEL ENGINEERING COLLEGE
(UNIT OF UDU)

APRIL 2025

**DEPARTMENT OF ELECTRONICS ENGINEERING
MODEL ENGINEERING COLLEGE, THRIKKAKARA
ERNAKULAM**

MODEL ENGINEERING COLLEGE, THRIKKAKARA



DEPARTMENT OF ELECTRONICS ENGINEERING

CERTIFICATE

*This is to certify that the project report entitled “**PROJECT NAME**” is a bonafide record of the project work done by **YOUR NAME** (VIII Semester B.Tech, Reg No. 13123XXX) towards the partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Electronics & Communication Engineering of APJ Abdul Kalam Technological University during the year 2025.*

PROJECT CO-ORDINATOR

PROJECT GUIDE

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ACKNOWLEDGEMENT

*First of all, I would like to thank the **Lord Almighty** who helped me to finish this project on time.*

I express my sincere thanks to , The Principal, Model Engineering College, Thrikkakara, for providing opportunity and the environment to do the project in my college.

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An excellent group of teaching and non-teaching staff helped me for this project. I owe much the assistance they gave me while doing the project.

Last, but not least I would like to thank my parents and friends for all the moral support and that they have given me.

Your Name (Roll No.)

ABSTRACT

Paste your abstract here

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Chapter 1

Latex

This is to for a chapter (Ch)

1.1 Section Name

This is to for a Section(1.1,2.1...)

1. First point.
2. Second point.

1.1.1 Subsection Name

This is to for a Subsection(1.1.1,2.3.1...)

- First point.
- Second point.

Chapter 2

Second Chapter

2.0.1 Equations & Equation arrays

$$w_k = \begin{cases} 0 & \tilde{c}_{i,j} = 2 \times Q \times \text{round}\left(\frac{c_{i,j}}{Q}\right) \\ 1 & \tilde{c}_{i,j} = 2 \times Q \times \text{round}\left(\frac{c_{i,j}-1}{Q}\right) + Q \end{cases} \quad (2.1)$$

The Equation 2.1 is above

$$Y = 0.299R + 0.587G + 0.114B \quad (2.2)$$

$$C_b = -0.1687R - 0.3313G - 0.5B + 128 \quad (2.3)$$

$$C_r = 0.5R - 0.4187G - 0.0813B + 128 \quad (2.4)$$

Chapter 3

Third Chapter

3.1 Figure



Figure 3.1: Logo1

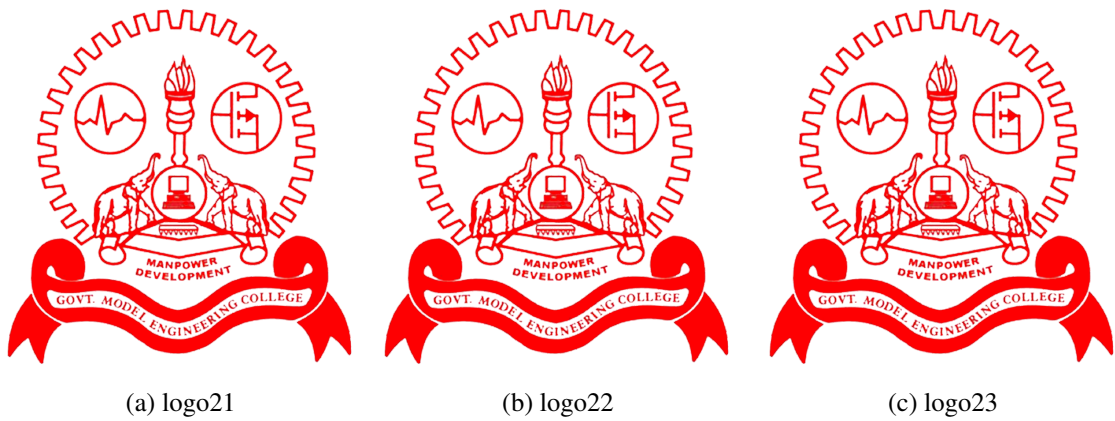


Figure 3.2: logo2

Chapter 4

Forth Chapter

4.1 Algorithm/Tcolorbox

ALGORITHM: AN IMAGE AUTHENTICATION & RECONSTRUCTION SCHEME

Require: I

Require: $h(), f(), g(), f^{-1}(), g^{-1}()$

Require: $b, B : b \leq B$

for $i = 1 \rightarrow N$ **do**

Reconstruct $I_i : e_i \neq 1$

Chapter 5

Fifth Chapter

5.1 Tables

No	Particular	Quantity	Unit Price	Amount
1	PIC 16F877A	1	150	150
2	Transformer	1	100	100
Total				1302

Conclusion

References & Links

- [1] Authors., "*Journal Name*", IEEE Transactions on Image Processing, vol.VolNo, pp.PageNo, Month & Year.
- [2] Author, *Book Name*:Publication, Year.
- [3] *Plasma Weapon - Wikipedia, the free encyclopedia*,
http://en.wikipedia.org/wiki/Plasma_Weapon

Appendices

Appendix A

Coding

```
clc;
close all;
clear all;
n1=input('Enter Starting Point n1: ');
n2=input('Enter End Point n2: ');
n=-n1:1:n2;

% Impulse Signal
amp_impl=[zeros(1,n1) 1 zeros(1,n2)];
figure(1);
subplot(221);
plot(n,amp_impl);
xlabel('Time');
ylabel('Amplitude');
title('Continious Impulse Signal');
subplot(222);
stem(n,amp_impl);
xlabel('Samples');
ylabel('Amplitude');
title('Discrete Impulse Signal');

% Step Signal
amp_stp=[zeros(1,n1) ones(1,n2+1)];
```

```
subplot(223);
stairs(n, amp_stp);
xlabel('Time');
ylabel('Amplitude');
title('Continious Step Signal');
subplot(224);
stem(n, amp_stp);
xlabel('Samples');
ylabel('Amplitude');
title('Discrete Step Signal');
```

```
% Ramp Signal
amp_rmp=[zeros(1,n1),n(n1+1:end)];
figure(2);
subplot(221);
plot(n, amp_rmp);
xlabel('Time');
ylabel('Amplitude');
title('Continious Ramp Signal');
subplot(222);
stem(n, amp_rmp);
xlabel('Samples');
ylabel('Amplitude');
title('Discrete Ramp Signal');
```

```
% Exponential Signal
n0=0:1:n2;
Am_exp=input('Enter Amplitude of Exponential Wave Am: ');
b=input('Enter Decaying Factor of Exponential Wave b: ');
```

```
amp_expl=Am_exp*exp(-b.*n0);
figure(2);
subplot(223);
plot(n0,amp_expl);
xlabel('Time');
ylabel('Amplitude');
title('Continious Exponential Signal');
subplot(224);
stem(n0,amp_expl);
xlabel('Samples');
ylabel('Amplitude');
title('Discrete Exponential Signal');

% Sine Wave
n=0:.02:1;
Am_sin=input('Enter Amplitude of Sine Wave Am: ');
f=input('Enter Frequency of Sine Wave f: ');
amp_sine=Am_sin*sin(2*pi*f*n);
figure(3)
subplot(221);
plot(n,amp_sine);
xlabel('Time');
ylabel('Amplitude');
title('Continious Sine Signal');
subplot(222);
stem(n,amp_sine);
xlabel('Samples');
ylabel('Amplitude');
title('Discrete Sine Signal');
```

```
% Square Wave
Am_sqre=input('Enter Amplitude of Square Wave Am: ');
f=input('Enter Frequency of Square Wave f: ');
dty=input('Enter Duty Cycle of Square Wave: ');
amp_sqre=Am_sqre*square(2*pi*f*n,dty);
subplot(223);
stairs(n,amp_sqre);
xlabel('Time');
ylabel('Amplitude');
title('Continious Square Signal');
subplot(224);
stem(n,amp_sqre);
xlabel('Samples');
ylabel('Amplitude');
title('Discrete Square Signal');

% Sawtooth Wave
Am_st=input('Enter Amplitude of Sawtooth Wave Am: ');
f=input('Enter Frequency of Sawtooth Wave f: ');
dty=input('Enter Duty Cycle of Sawtooth Wave: ');
amp_st=Am_st*sawtooth(2*pi*f*n,dty);
figure(4);
subplot(221);
plot(n,amp_st);
xlabel('Time');
ylabel('Amplitude');
title('Continious Sawtooth Signal');
subplot(222);
```

```
stem(n, amp_st);  
xlabel('Samples');  
ylabel('Amplitude');  
title('Discrete Sawtooth Signal');  
  
% Pulse Signal  
n01=input('No. of zeros left to gate signal n01: ');  
n11=input('No. of ones n11: ');  
n02=input('No. of zeros right to gate signal n02: ');  
n_sam=n01+n11+n02;  
n=0:1:n_sam-1;  
amp_gt=[zeros(1,n01) ones(1,n11) zeros(1,n02)];  
subplot(223);  
stairs(n, amp_gt);  
xlabel('Time');  
ylabel('Amplitude');  
title('Continious Gate Signal');  
subplot(224);  
stem(n, amp_gt);  
xlabel('Samples');  
ylabel('Amplitude');  
title('Discrete Gate Signal');
```

```
clc;
close all;
clear all;
n_st=input('Enter the Starting Point of the Sequence; n = ');
x=input('Enter the Sequence: x(n) = ');
x_len=length(x);
n_ed=n_st+x_len-1;
n=n_st:n_ed;
figure (1);
subplot(3,1,1);
stem(n,x);
xlabel('Samples');
ylabel('Amplitude');
title('Time Shifting');

% Time Shifting (1st method)
k=input('Enter the shifting factor: k = ');
sh_sq01=n+k;
subplot(3,1,2);
stem(sh_sq01,x);
xlabel('Samples');
ylabel('Amplitude');

% Time Shifting (2nd method)
n1=min(n_st,n_st+k);
n2=max(n_ed,n_ed+k);
n_sh=n1:n2;
sh_sq02=[zeros(1,k) x zeros(1,-k)];
subplot(3,1,3);
```

```
stem(n_sh , sh_sq02 );
xlabel( ' Samples ' );
ylabel( ' Amplitude ' );

% Time Scaling
a=input('Enter the scaling factor: a = ');
figure (2)
subplot(311);
stem(n,x);
xlabel( ' Samples ' );
ylabel( ' Amplitude ' );
if a > 1
    b=mod(n,a);
    c=[];
    d=[];
    for i=1:x_len
        if b(i)==0
            c=x(i);
            d=[d c];
        end
    end
    d;
    n_start=ceil(n_st/a);
    n_end=floor(n_ed/a);
    n_sc=n_start:n_end;
else
    e=(x_len/a)-(1/a)+1;
    d=[zeros(1,e)];
```



```
d(1:1/a:end)=x;
n_start=n_st/a;
n_end=n_ed/a;
n_sc=n_start:n_end;
end
subplot(312);
stem(n_sc,d);
xlabel('Samples');
ylabel('Amplitude');
title('Time Scaling');

% Time Reversal
x;
n;
nrev=-n;
x_rev=x(end:-1:1);
n_rev=nrev(end:-1:1);
subplot(313);
stem(n_rev,x_rev);
xlabel('Samples');
ylabel('Amplitude');
title('Time Reversal');
```

Appendix B

Project Estimate

Appendix C

Datasheets